MANASVI SAXENA

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SUMMARY

PhD in Computer Science (UIUC, 2024) focused on formal methods, runtime verification, and semantics-based reasoning. Direct experience with Amazon's S3 Automated Reasoning Group across two summers as Applied Scientist intern.

EDUCATION

Ph.D. Computer Science, University of Illinois Urbana-Champaign Dec 2024

- Advisor - Prof. Grigore Roşu
- Thesis - A Semantics-First Approach to Safe Guidelines-Based Clinical Decision Support
- Designed MediK, a domain-specific language with formally defined K-Framework semantics; derived execution and analysis tools correct-by-construction; verified safety properties of a real-world clinical decision support system.

M.S. Computer Science, University of Illinois Urbana-Champaign May 2018

- Relevant Coursework - Formal Methods of Software Development, Runtime Verification, Logic in Computer Science, Computer and Cryptocurrency Security

B.S. Computer Science, University of Illinois Urbana-Champaign May 2015

- Relevant Coursework - Programming Languages and Compilers, Fundamental Algorithms, Theory of Computation, Computer Networking, Artificial Intelligence

EXPERIENCE

Tarnyq, Bengaluru, India February 2026 - Present
Co-Founder *Bengaluru, India*

- Co-founded a venture (`https://tarnyq.net`) focused on software for safety-critical settings, using formal-methods, semantics-first techniques.
- Extending thesis work on MediK, a domain-specific language for expressing medical guidelines with formal executable semantics, toward intelligent, trustworthy decision support with formally verified correctness properties. Developing conformance monitoring solutions to safely integrate AI by ensuring AI-generated advice remains guideline-compliant.
- Building software for safety-critical settings that is secure by construction rather than retrofitted: end-to-end encryption with keys held on-device, distributed replication for resilient backup and offline use, and verifiable-integrity techniques inspired by blockchain systems.

Satyax Inc., Urbana, Illinois January 2025 - February 2026
Founder & CEO *Urbana, IL*

- Scaling doctoral research on formally verified clinical decision support from research prototype to production, applied initially to acute conditions such as sepsis with formal-methods-backed correctness properties.
- Adapting the platform for real-world clinical use: integrating with patient monitors, automating clinical workflows, and surfacing richer patient context to clinicians.

S3 Automated Reasoning Group, Amazon Web Services (AWS) Summer 2021 & 2020
Applied Scientist Intern

- Developed techniques and tooling for evaluating compliance of highly concurrent, distributed code against relevant properties expressed as RV-Monitor specifications (`https://github.com/runtimeverification/rv-monitor`) utilizing the existing test suite.
- Optimized RV-Monitor, an academic tool, to handle millions of lines of code, and enabled support for expressing properties relevant to concurrent systems, such as read-after-write consistency and serializability efficiently, enabling RV-Monitor to handle the scale of massively parallel and highly distributed systems like S3.

Formal Systems Laboratory, University of Illinois May 2016 - December 2024
Research Assistant *Urbana, IL*

- Developed the MediK Framework, a DSL for encoding medical guidelines using state machines with strong correctness guarantees (`https://github.com/medik-framework`).
- Maintained RV-Monitor, a language-independent monitoring framework that supports rich formalisms for expressing properties.
- Contributed to KEVM, formalizing the semantics of the Ethereum Virtual Machine (EVM) in the K Framework (`http://github.com/runtimeverification/evm-semantics`).

Runtime Verification Inc., Urbana, Illinois July 2015 - May 2016
Formal Verification Engineer *Urbana, IL*

- Worked on a team developing a dynamic analysis tool for finding undefined behavior in C programs, using formal semantics of C.
- Worked on improving tool's performance, and evaluated its applications on large codebases.

PROJECTS

The MediK Framework - Trustworthy Intelligent Decision Support

- *Domain-specific language* for encoding medical guidelines as verifiable state machines with formal K-Framework semantics.
- *Safety guarantees* established through formal analysis with semantics-derived, correct-by-construction tools, while maintaining clinical interpretability.
- *Comprehensive approach* combines formal reasoning with clinician-validated clinical knowledge.
- *Repository:* `https://github.com/medik-framework`

RV-Monitor

- *Core team member* maintaining language-agnostic monitor framework.
- *Optimized for enterprise-scale systems* with *rich formalisms* for expressing complex properties.
- *Repository:* `https://github.com/runtimeverification/rv-monitor`

KEVM - Semantics of the EVM in K

- *Formalization* of the Ethereum Virtual Machine (EVM) in the K Framework. Contributed to the semantics of the Ethereum ABI, as an extension to the original semantics.
- *Enables formal analysis* of smart contracts with high degree of automation.
- *Repository:* `https://github.com/runtimeverification/evm-semantics`

CORE COMPETENCIES

Automated Reasoning:	Formal Semantics, Symbolic Execution, SMT Solvers, Program Analysis, Model Checking, Deductive Verification, Runtime Monitoring, Testing
Programming:	Java, Haskell, K Framework, Scala, Python, C

PUBLICATIONS

- Manasvi Saxena. *A semantics-first approach to safe guidelines-based clinical decision support*. PhD thesis, University of Illinois Urbana-Champaign, December 2024. Available at `https://hdl.handle.net/2142/127206`
- Manasvi Saxena, Shuang Song, and Lui Sha. MediK: Towards Safe Guideline-based Clinical Decision Support. In *2023 Formal Methods in Computer-Aided Design (FMCAD)*, pages 306–317, 2023
- Everett Hildenbrandt, Manasvi Saxena, Nishant Rodrigues, Xiaoran Zhu, Philip Daian, Dwight Guth, Brandon Moore, Daejun Park, Yi Zhang, Andrei Stefanescu, et al. KEVM: A complete formal semantics of the Ethereum Virtual Machine. In *2018 IEEE 31st Computer Security Foundations Symposium (CSF)*, pages 204–217. IEEE, 2018
- Dwight Guth, Chris Hathhorn, Manasvi Saxena, and Grigore Rosu. Rv-match: Practical semantics-based program analysis. In *Computer Aided Verification - 28th International Conference, CAV 2016, Toronto, ON, Canada, July 17-23, 2016, Proceedings, Part I*, volume 9779 of *LNCS*, pages 447–453. Springer, July 2016

- Daejun Park, Yi Zhang, Manasvi Saxena, Philip Daian, and Grigore Rosu. A formal verification tool for ethereum VM bytecode. In Gary T. Leavens, Alessandro Garcia, and Corina S. Pasareanu, editors, *Proceedings of the 2018 ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, ESEC/SIGSOFT FSE 2018, Lake Buena Vista, FL, USA, November 04-09, 2018*, pages 912–915. ACM, 2018
- Shuang Song, Manasvi Saxena, Pei-Hsuan Tsai, and Lui Sha. Towards modular and formally-verifiable software architecture for clinical guidance systems. In *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pages 4271–4276. IEEE, 2023
- Philip Daian, Dwight Guth, Chris Hathhorn, Yilong Li, Edgar Pek, Manasvi Saxena, Traian Florin Serbanuta, and Grigore Rosu. Runtime verification at work: A tutorial. In *Runtime Verification - 16th International Conference, RV 2016 Madrid, Spain, September 23-30, 2016, Proceedings*, volume 10012 of *Lecture Notes in Computer Science*, pages 46–67. Springer, September 2016